

Python: Fixing Common Errors

When your code does not run, it is almost always one of these small things. Check this list first!

✓ Error 1: Indentation (the spaces in front of a line)

Python uses **spaces in front of a line** to know what belongs inside a block.

These three things always need the lines under them to be **indented** (pushed in):

- `def` (a function)
- `if`
- `while` / `for` loops

✗ Wrong (not indented)

```
def on_chat():  
forward = 5
```

✓ Right (indented)

```
def on_chat():  
    forward = 5
```

👉 The line `forward = 5` is **inside** the function, so it must be pushed in.

How to indent: - Press **Enter** to make a new line. - Press **Tab** to push the line in (indent). - Press **Backspace** to pull the line back out.

✓ **Error 2: Don't forget the `:` (colon)**

`def`, `if`, `while`, and `for` lines must **end with a colon** `:`

✗ **Wrong**

```
while True
if forward == 5
```

✓ **Right**

```
while True:
if forward == 5:
```

👉 No colon = error. Always check the end of the line.

✓ **Error 3: Keep the function name on the same line as** `def`

The function name goes **right after** `def`, **on the same line**, ending with `()` and `:`

✗ **Wrong**

```
def  
on_chat():
```

✓ **Right**

```
def on_chat():
```

✓ Error 4: Match your indentation

Lines inside the **same block** must line up with the same number of spaces.

✗ Wrong (lines don't line up)

```
def on_chat():  
    forward = 5  
    agent.move(FORWARD, forward)
```

✓ Right (lines line up)

```
def on_chat():  
    forward = 5  
    agent.move(FORWARD, forward)
```

👉 A loop inside a function gets indented **again**:

```
def on_chat():  
    for layer in range(3):  
        agent.move(FORWARD, 5)
```

Practice Exercises

Fix each one. The error is written above it.

Exercise 1 — Missing colon

```
def on_chat()  
    agent.move(FORWARD, 3)  
player.on_chat("go", on_chat)
```

Exercise 2 — Not indented

```
def on_chat():  
agent.move(UP, 1)  
player.on_chat("up", on_chat)
```

Exercise 3 — Missing colon on the loop

```
def on_chat():
    for i in range(4)
        agent.move(FORWARD, 1)
        agent.turn(TurnDirection.LEFT)
player.on_chat("square", on_chat)
```

Exercise 4 — Wrong indentation (lines don't line up)

```
def on_chat():
    forward = 5
    agent.move(FORWARD, forward)
player.on_chat("run", on_chat)
```

Exercise 5 — Loop not indented inside the function

```
def on_chat():
for layer in range(3):
    agent.move(UP, 1)
player.on_chat("tower", on_chat)
```



Answer Key

Exercise 1

```
def on_chat():
    agent.move(FORWARD, 3)
player.on_chat("go", on_chat)
```

Exercise 2

```
def on_chat():
    agent.move(UP, 1)
player.on_chat("up", on_chat)
```

Exercise 3

```
def on_chat():
    for i in range(4):
        agent.move(FORWARD, 1)
        agent.turn(TurnDirection.LEFT)
player.on_chat("square", on_chat)
```

Exercise 4

```
def on_chat():  
    forward = 5  
    agent.move(FORWARD, forward)  
player.on_chat("run", on_chat)
```

Exercise 5

```
def on_chat():  
    for layer in range(3):  
        agent.move(UP, 1)  
player.on_chat("tower", on_chat)
```

Remember the 4 checks

1. **Indent** the lines inside `def`, `if`, `while`, `for` (press Tab).
2. End `def` / `if` / `while` / `for` lines with `:`
3. Keep the function name **next to** `def`.
4. Make lines in the same block **line up**.

Press **Enter** for a new line, **Tab** to indent, **Backspace** to go back.